

Mining Multiple-Level Frequent Patterns

- Items often form hierarchies

- Milk: Dairyland 2% milk
- Bread: Wonder wheat bread

Uniform support

Level 1
min_sup = 5%

Level 2
min_sup = 5%

Milk
[support = 10%]

2% Milk
[support = 6%]

Skim Milk
[support = 2%]

Reduced support

Level 1
min_sup = 5%

Level 2
min_sup = 1%

- Min-support thresholds for items following concept hierarchies

- **Uniform** min-support across multiple levels (reasonable?)
- **Level-reduced** min-support:
 - Items at the lower level are expected to have lower support

- Efficient mining: *Shared* multi-level mining

- Use the lowest min-support to pass down the set of candidates

Redundancy Filtering at Mining Multi-Level Associations

- ❑ Redundancy filtering: redundant due to “ancestor” relationships
 - ❑ milk $\Rightarrow$ wheat bread [support = 8%, confidence = 70%] (1)
 - ❑ 2% milk $\Rightarrow$ wheat bread [support = 2%, confidence = 72%] (2)
 - ❑ Suppose the 2% milk sold is about $\frac{1}{4}$ of milk sold in gallons
 - ❑ (2) should be able to be “derived” from (1)
- ❑ **In general:** A rule is *redundant* if its support is close to the “expected” value, according to its “ancestor” rule, and it has a similar confidence as its “ancestor”

Q: If rule (1) is an ancestor of rule (2), which one to prune?

Customized Min-Supports for Different Kinds of Items

- ❑ We have used the same min-support threshold for all the items or item sets to be mined in each association mining
- ❑ In reality, some items (e.g., diamond, watch, ...) are valuable but less frequent
- ❑ It is necessary to have customized min-support settings for different kinds of items
- ❑ One Method: Use **group-based “individualized” min-support**
 - ❑ E.g., {diamond, watch}: 0.05%; {bread, milk}: 5%; ...
 - ❑ How to mine such rules efficiently?
 - ❑ Existing scalable mining algorithms can be easily extended to cover such cases